

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Develop a Student Profile Generator using JavaScript ES6 Classes and DOM Manipulation

2. AIM

To implement a dynamic web application that allows users to create student profiles using JavaScript ES6 classes and dynamically render them as styled cards by manipulating the Document Object Model (DOM).

3. OBJECTIVE

- To understand and declare ES6 Class structures with constructor methods.
- To handle click events dynamically using `addEventListener` callbacks.
- To extract text and number values from form inputs in real-time.
- To construct and insert new DOM elements dynamically on user interactions.
- To style generated profile containers with gradients, borders, and shadows.

4. THEORY

JavaScript ES6 classes introduce a cleaner object-oriented syntax over prototypes. A class is defined using the `class` keyword and instantiated using the `new` keyword. The constructor method initializes properties on the instance.

DOM (Document Object Model) manipulation allows JavaScript to update web pages in real-time. Methods like `document.createElement()` generate new nodes, `textContent` and `innerHTML` inject value content, and `appendChild()` updates the document tree by nesting child elements under a parent element.

5. ALGORITHM / PROCEDURE

1. Create an HTML template file named `student_profile.html`.
2. Configure input containers for Name, Roll No, Department, and CGPA, along with a submit button.
3. Write CSS styling for a background gradient and a clean card grid.

4. Configure input focus styles to turn input borders green on active state.
5. Define the Student class with a constructor assigning fields to instance properties.
6. Retrieve the profile parent element and clear previous cards on click.
7. Attach a click handler to the creation button.
8. Inside the click callback, read input box values.
9. Instantiate a Student object with these values.
10. Dynamically create card containers, header, and paragraph elements.
11. Inject student attributes using innerHTML and textContent.
12. Append the paragraphs to the card container, then append the card to the profile parent element.
13. Save and verify profile creation in a web browser.

6. PROGRAM

student_profile.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile</title>
  <style>
    body { font-family: Arial, sans-serif; background: linear-gradient(135deg,
lightblue, pink, lightgreen); text-align: center; padding: 20px; margin: 0; }
    h1 { color: #234; }
    .form-container { background-color: rgba(255, 255, 255, 0.9); width: 320px;
margin: 20px auto; padding: 20px; border-radius: 15px; box-shadow: 0 5px 15px rgba(0, 0,
0, 0.2); }
    input { width: 90%; padding: 10px; margin: 8px 0; border: 2px solid #9ad9e8;
border-radius: 8px; font-size: 14px; box-sizing: border-box; }
    input:focus { outline: none; border-color: #66bb6a; }
    button { padding: 10px 20px; background-color: #66bb6a; color: white; border:
none; border-radius: 8px; font-size: 15px; cursor: pointer; }
    button:hover { background-color: #43a047; }
    #profile { width: 360px; margin: 20px auto; }
    .student-card { background: linear-gradient(135deg, #e3f2fd, #fce4ec, #e8f5e9);
padding: 20px; border-radius: 15px; box-shadow: 0 5px 15px rgba(0, 0, 0, 0.2); text-align:
left; border-left: 6px solid #66bb6a; }
    .student-card h2 { text-align: center; color: #1976d2; margin-bottom: 15px; font-
size: 20px; }
    .student-card p { font-size: 15px; margin: 10px 0; color: #333; }
    .student-card p strong { color: #388e3c; }
  </style>
</head>
<body>
  <h1>Student Profile Generator</h1>
  <div class="form-container">
    <input type="text" id="name" placeholder="Enter Name" value="Rajesh">
    <input type="text" id="rollNo" placeholder="Enter Roll Number" value="23A91A0501">
    <input type="text" id="department" placeholder="Enter Department" value="CSE">
    <input type="number" id="cgpa" placeholder="Enter CGPA" step="0.01" value="9.25">
```

```

        <br><br>
        <button id="createBtn">Create Student Profile</button>
    </div>
    <div id="profile"></div>
    <script>
        class Student {
            constructor(name, rollNo, department, cgpa) {
                this.name = name;
                this.rollNo = rollNo;
                this.department = department;
                this.cgpa = cgpa;
            }
        }
        document.getElementById("createBtn").addEventListener("click", function() {
            let name = document.getElementById("name").value;
            let rollNo = document.getElementById("rollNo").value;
            let department = document.getElementById("department").value;
            let cgpa = document.getElementById("cgpa").value;
            let student = new Student(name, rollNo, department, cgpa);
            let profile = document.getElementById("profile");
            profile.innerHTML = "";
            let card = document.createElement("div");
            card.className = "student-card";
            let heading = document.createElement("h2");
            heading.textContent = "Student Profile";
            let nameElement = document.createElement("p");
            nameElement.innerHTML = "<strong>Name:</strong> " + student.name;
            let rollElement = document.createElement("p");
            rollElement.innerHTML = "<strong>Roll No:</strong> " + student.rollNo;
            let departmentElement = document.createElement("p");
            departmentElement.innerHTML = "<strong>Department:</strong> " +
student.department;
            let cgpaElement = document.createElement("p");
            cgpaElement.innerHTML = "<strong>CGPA:</strong> " + student.cgpa;
            card.appendChild(heading);
            card.appendChild(nameElement);
            card.appendChild(rollElement);
            card.appendChild(departmentElement);
            card.appendChild(cgpaElement);
            profile.appendChild(card);
        });

        // Auto trigger for screenshot 2
        if (window.location.search.includes("render=true")) {
            window.addEventListener('DOMContentLoaded', () => {
                document.getElementById("createBtn").click();
            });
        }
    </script>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<code>class Student { ... }</code>	Declares the ES6 Class model to structure Student object properties.
<code>constructor(name, rollNo, ...)</code>	Method that runs automatically when a student object is instantiated, setting properties.
<code>addEventListener('click', ...)</code>	Binds a function execution callback to click events on the submit button.
<code>document.getElementById(...)</code>	Finds and returns HTML DOM elements by matching their unique ID attributes.
<code>new Student(name, rollNo, ...)</code>	Instantiates a new instance of the Student class with form input values.
<code>document.createElement('div')</code>	Generates a new empty div element node inside the DOM structure.
<code>card.className = 'student-card'</code>	Assigns a style class name to configure the CSS layout of the generated card.
<code>nameElement.innerHTML = ...</code>	Modifies the inner markup of a paragraph tag to format label strings in bold.
<code>card.appendChild(heading)</code>	Appends the heading node to the card container element tree.
<code>profile.appendChild(card)</code>	Inserts the final student card container inside the profile display div.

8. TEST CASES AND EXECUTION DATA

Test Case	Name	Roll No	Department	CGPA	Expected Result	Actual Result	Status
TC-01	Rajesh	23A91A0501	CSE	9.25	Generates student profile card showing details	Displayed correctly	Pass
TC-02	Purna	23A91A0502	AI & ML	8.80	Generates student	Displayed correctly	Pass

					profile card showing details		
TC-03	Sandeep	23A91A0503	ECE	7.50	Generates student profile card showing details	Displayed correctly	Pass
TC-04					Generates card displaying empty value spaces	Displayed correctly	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-05	Click inputs to focus	Selected field border turns green (#66bb6a) indicating active state	Pass

9. OUTPUT

Output 1: Initial Blank Profile Generator

Student Profile Generator

Create Student Profile

Output 3: Student Profile Card Generated

Student Profile Generator

Create Student Profile

Student Profile

Name: Rajesh
Roll No: 23A91A0501
Department: CSE
GPA: 9.25

10. RESULT

Thus, an object-oriented Student Profile Generator page was successfully developed using JS ES6 classes and DOM insertion methods, allowing dynamic student records to be created and displayed as card views.